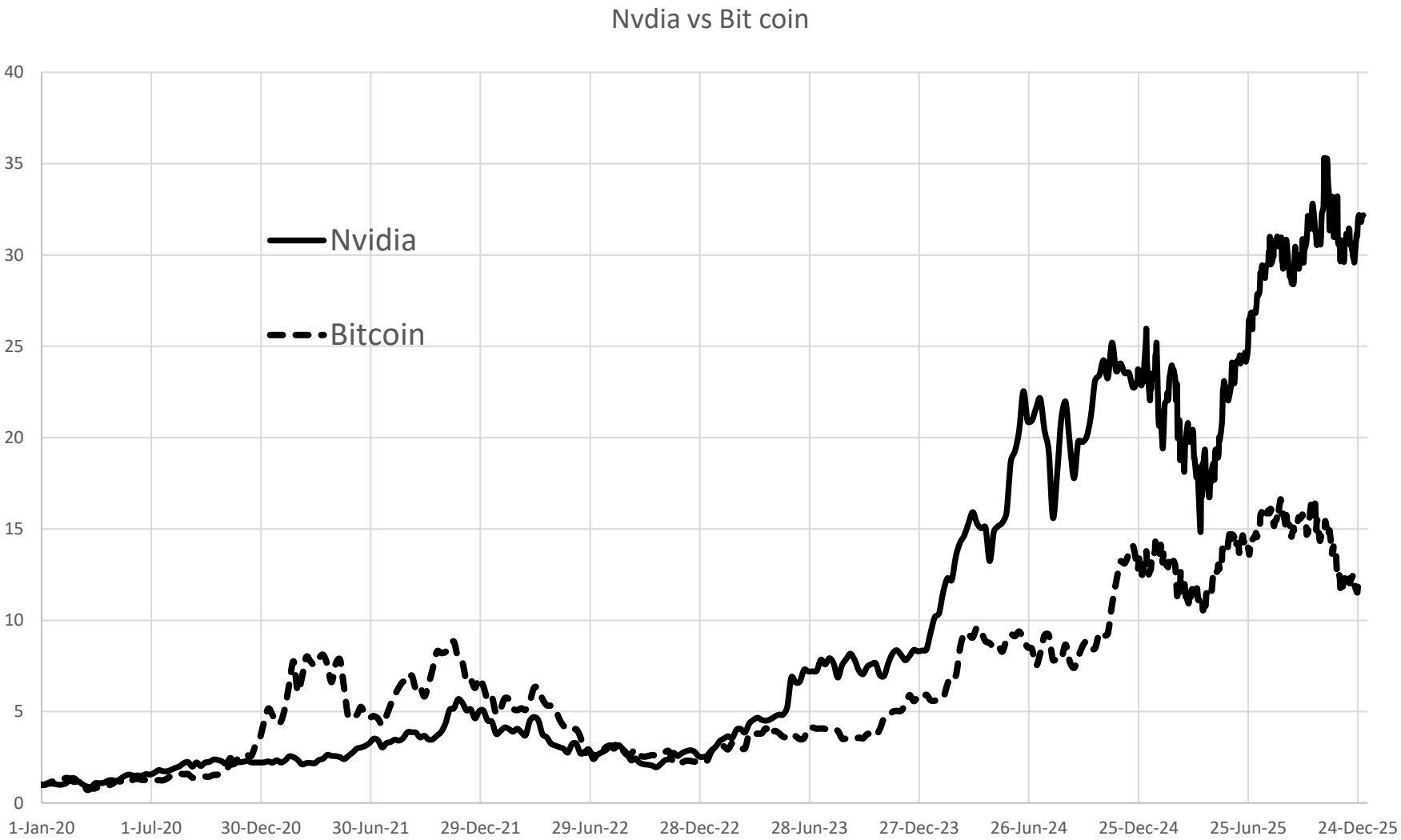
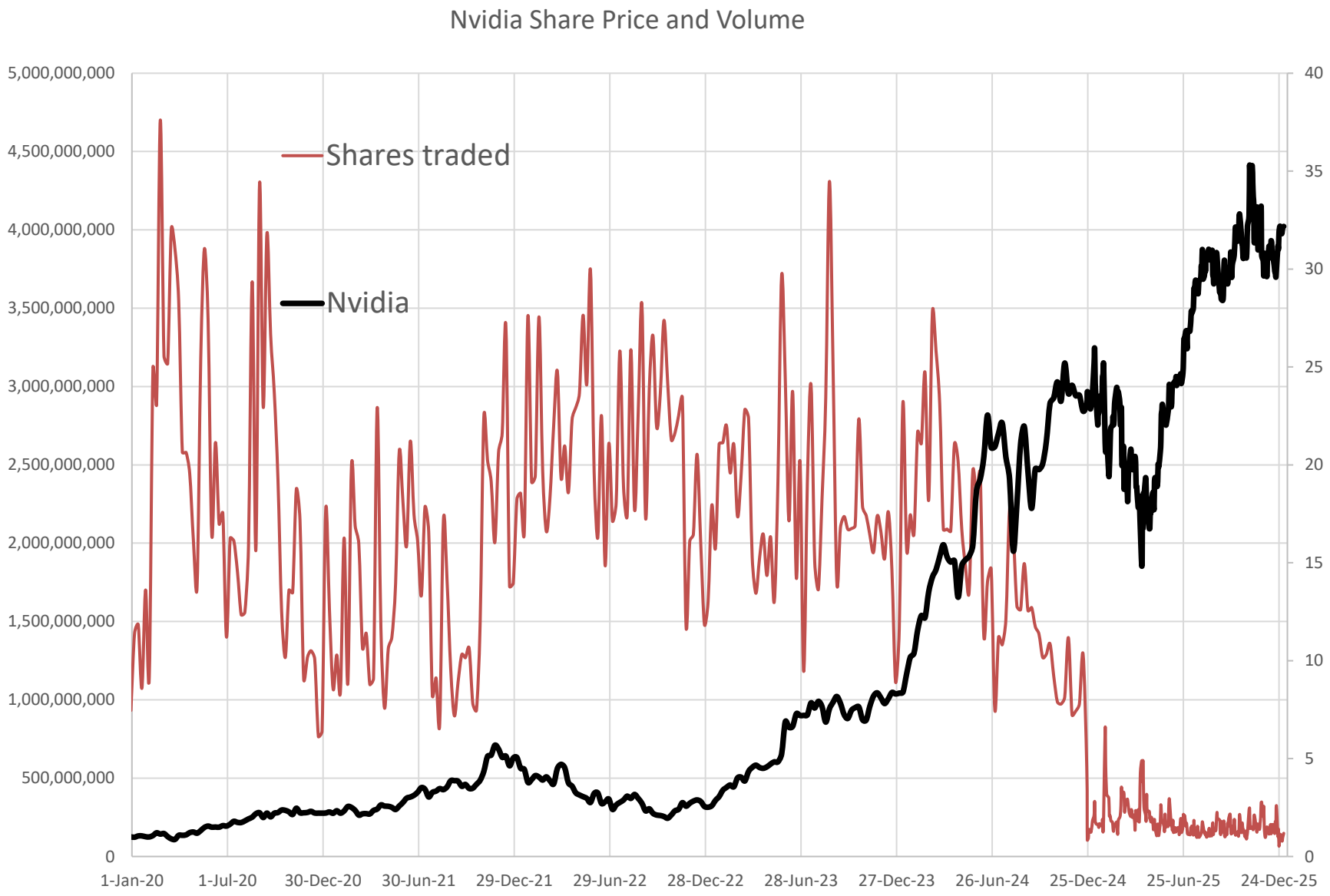

Nvdia vs Bit coin
Nvidia
Bitcoin
40
35
30
25
20
15
10
5
0
1-Jan-20
1-Jul-20
30-Dec-20
30-Jun-21
29-Dec-21
29-Jun-22
28-Dec-22
28-Jun-23
27-Dec-23
26-Jun-24
25-Dec-24
25-Jun-25
24-Dec-25


Nvidia Share Price and Volume
Shares traded
Nvidia
5,000,000,000
4,500,000,000
4,000,000,000
3,500,000,000
3,000,000,000
2,500,000,000
2,000,000,000
1,500,000,000
1,000,000,000
500,000,000
0
40
35
30
25
20
15
10
5
0
1-Jan-20
1-Jul-20
30-Dec-20
30-Jun-21
29-Dec-21
29-Jun-22
28-Dec-22
28-Jun-23
27-Dec-23
26-Jun-24
25-Dec-24
25-Jun-25
24-Dec-25

Class 1:
From claims to value

What is Finance?
Your Finance
My finance

A road map for the class
Objects of Study

What is Finance?

- The study of
  - Investors (who buy and hold claims and get resources later)
  - Issuers (who sell claims and use resources now)
  - Intermediaries who organize the
    - Creation, transfer, and valuation of claims
      - On market and not on market
    - Collection and dissemination of information
      - Public and Private

- The study and valuation of risk
  - An asset's return = riskless return + price of risk
- The study and valuation of liquidity
  - An asset's return = riskless return + price for illiquidity

- Not how to make money (it may be how to avoid losing it)

Finance is old

- We know of speculative crises under the Roman Empire
- We know of written claims in the fertile crescent on cuneiform tablets
- Around 1500 Philip II of Spain borrowed money with contingent repayment option depending on the arrival of the fleet and where lenders could convert their short term claims into long term bonds at pre specified prices
- The earliest join stock companies are from the same period (though there is a claim that the mills in Toulouse are a medieval traded corporation)

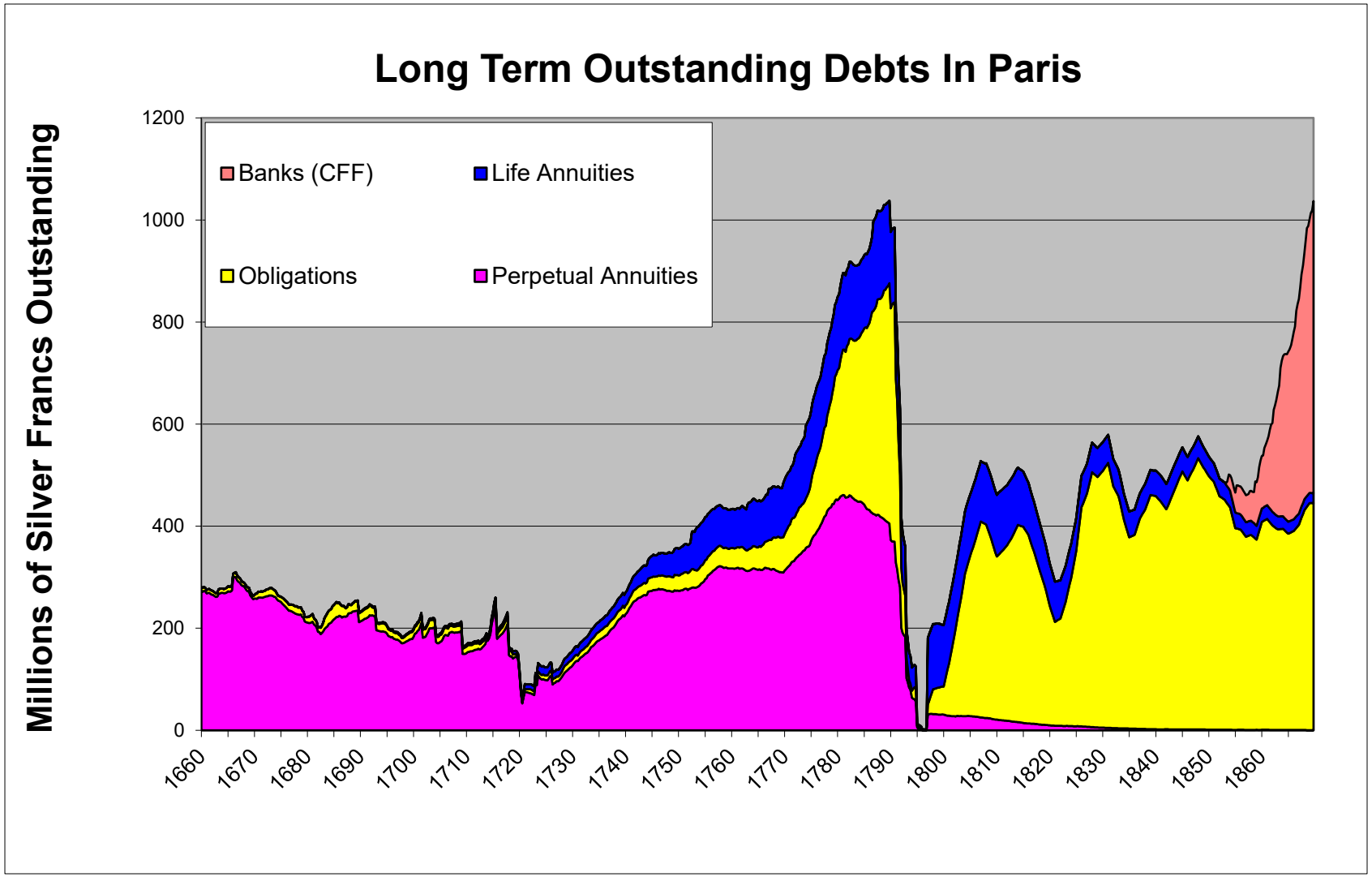
Finance is changing very fast

- Technology
  - Move to electronic markets
  - Attempts at crowd sourcing
- Data mining techniques and computer trading
- Changes in society are raising the demand for finance .
- Regulation

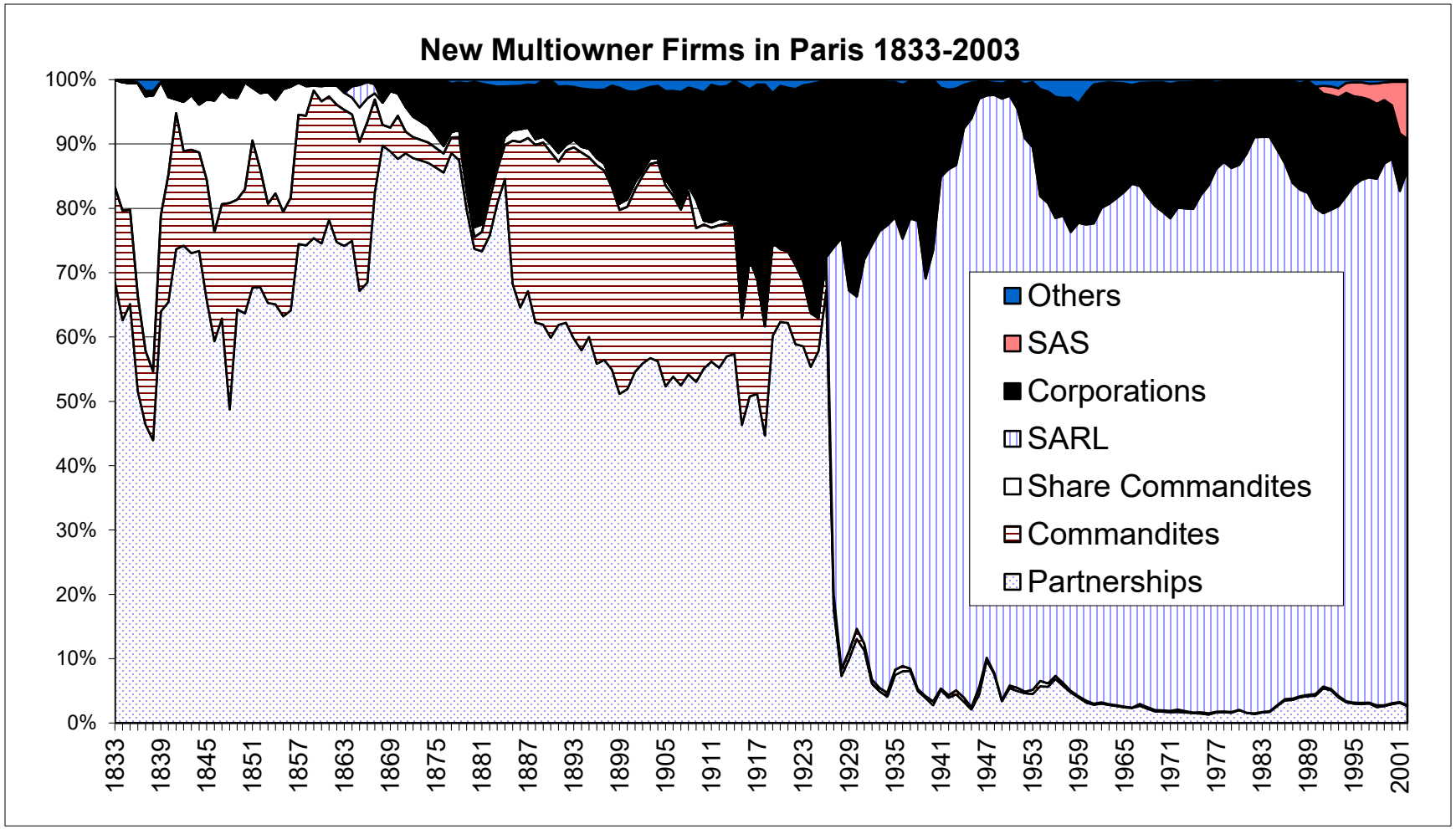
Your finance

- Why should you be in this class
- Most of you will be rich (I expect)
  - and you will save (I hope)
  - So some financial know how can be useful
- Some of you will be entrepreneurs or managers
- Some of you may work on wall street
- A very few of you may become financial economists
- For all these reasons finance matters to you.

My Finance (1)

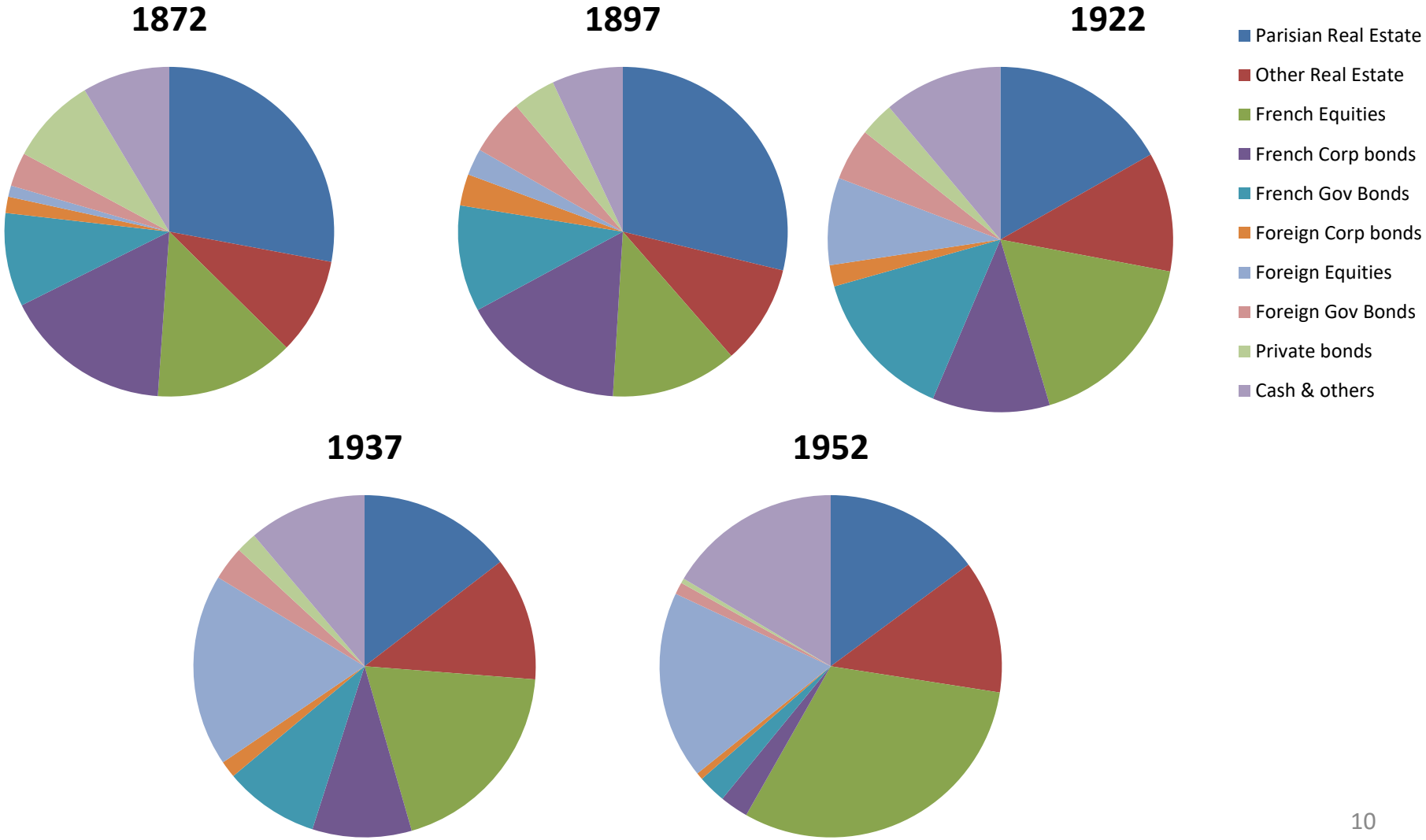

Long Term Outstanding Debts In Paris
Millions of Silver Francs Outstanding
Banks (CFF)
Life Annuities
Obligations
Perpetual Annuities
1200
1000
800
600
400
200
0
1660
1670
1680
1690
1700
1710
1720
1730
1740
1750
1760
1770
1780
1790
1800
1810
1820
1830
1840
1850
1860

My Finance (2)

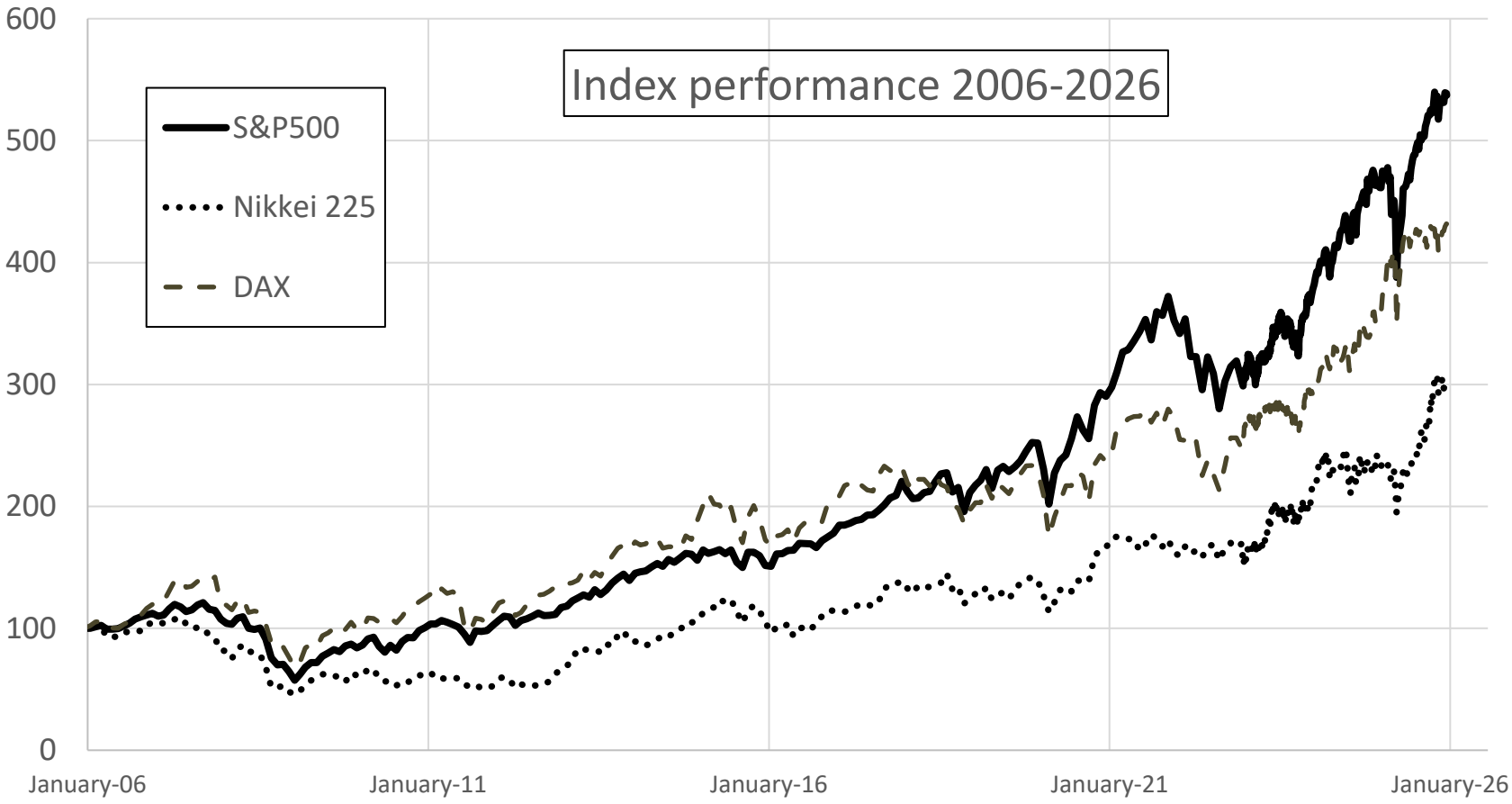

New Multiowner Firms in Paris 1833-2003
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
1833
1839
1845
1851
1857
1863
1869
1875
1881
1887
1893
1899
1905
1911
1917
1923
1929
1935
1941
1947
1953
1959
1965
1971
1977
1983
1989
1995
2001
Others
SAS
Corporations
SARL
Share Commandites
Commandites
Partnerships

My Finance (3)

Distributions of portfolios in Paris

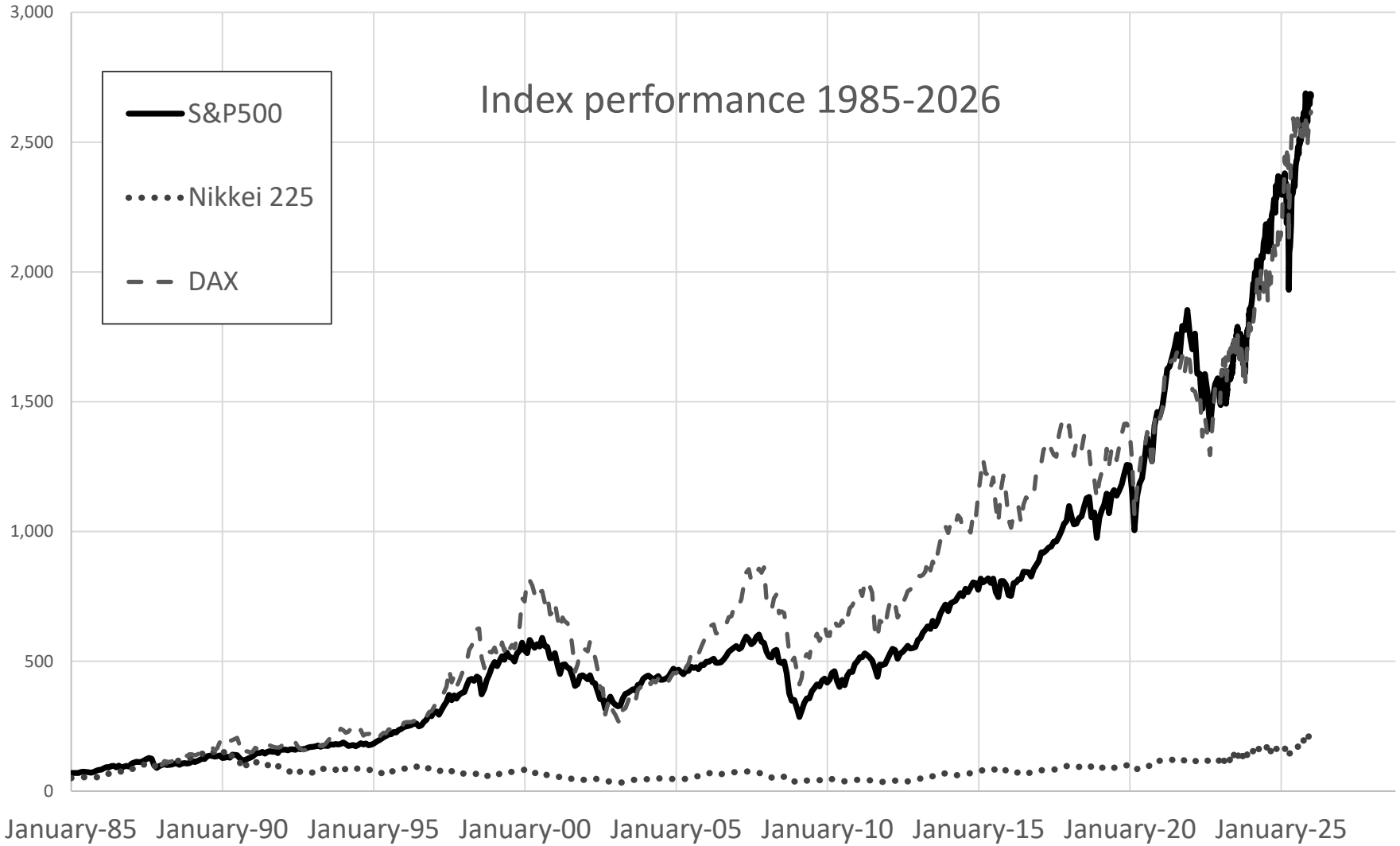


Your finance



Looking back

(JLR' graduation)



Goal for this class

- Financial Literacy
- An overview of the range of financial problems
  - Firms, individuals, governments intermediaries
- An introduction to financial problems
  - And how to solve them
- Close connection to tools of economics
  - Optimization
- Stay away from advanced techniques
  - Two states (good bad) Discrete time
- Other classes for more complicate (and realistic) setting. Here want to get the ideas clear.

A Road map for the Class

- Module 1: Finance literacy
- Module 2: Price of time
- Module 3-4: Price of risk and portfolios
- Module 5-6: beyond equity and bonds
- Module 7: corporate finance
- Module 8 : Crypto (one day)
- Three guest lectures to connect the material to what is out there.

How to go about it

- Each class
  - Some financial literacy
    - Today : what is an asset what is a claim
  - Some basic ‘theory’
    - Today: Present value, net present value, discounting and interest rates
  - Some empirical evidence about the theory
  - Some ways these theories are implemented in the ‘real’ world
- Homework
  - Deals with financial literacy and theory
  - Read the press, finance is now everywhere
- Finance journals
  - Deals with the real world.

Finance journals

- You will choose assets and evaluate their performance over the course of the class.
  - A real-world way to evaluate financial methods
  - And some consideration of the value and risks of stock picking.
    - Are you lucky or are you wise?
- Your return: extra credit.
  - Reward is scaled by the return to your choice divided by their variance. If you do not turn in a portfolio one week your return for that week's portfolio is automatically zero
- This week: pick a publicly traded bond.
- How to find one?
  - Good question, its part of the work.

Attendance

- Cost of attending Caltech $80,000+. In addition, if you were in the labor force income $50,000.
- If you take 4 classes each quarter you will have just shy of 360 hours of class time (cost per hour $500). Not attending seems like you are throwing money away.
- But somehow lecture classes are poorly attended.
  - So we have in-class quizzes, one a week and they cover the previous week's homework (so week 2 you are quizzed on the homework due that Monday) or the previous lecture.
  - You have to be in class to do the quiz.
- Ask questions!

Testing in the AI world

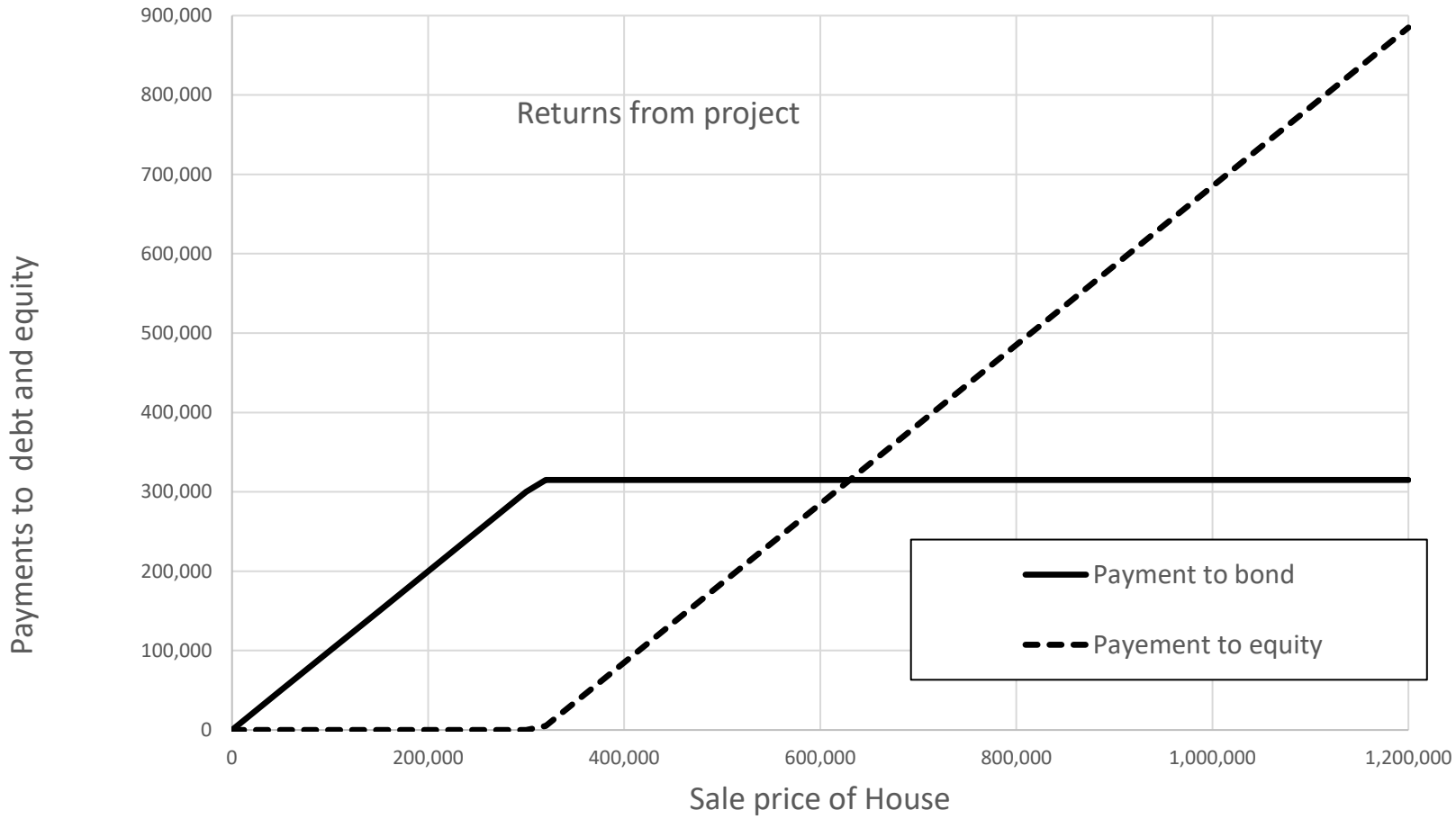
- Human capital has value because a computer cannot do it.
  - Its an open problem as to what that means in an AI world
  - This year new tack
    - Leave aside hard to compute problems—this is intro finance & AI is often better at this than either you or I.
    - Focus on problems where you confront yourself on your own. 1 hour mid term 1 hour final. 10 minutes recitation problem weeks 2-8

An introduction to Finance

- Assets
  - An asset is something that has value over time.
    - Cash flow
  - Real assets vs financial assets
- Economies are rich
  - When they have a lot of real assets AND these assets are used productively
  - Requires getting the assets to those who can used them productively.

A project

- Purchase a house ($500,000), refurbish it, sell.
  - House is an asset (produces a flow of rents)
- Many decisions focus on the finance
  - Decision to invest $150,000 in improvement
  - House ready to resell in 1 year
- Need to find $650,000 to make the project work.
  - Borrow $300,000 and have to pay lender 315,000 when house sells.


Returns from project
Payments to debt and equity
900,000
800,000
700,000
600,000
500,000
400,000
300,000
200,000
100,000
0
0
200,000
400,000
600,000
800,000
1,000,000
1,200,000
Sale price of House
Payment to bond
Payement to equity

Price of time, risk, liquidity

- Price of time (5%)
- Price of risk (not shown)
  - Bondholder can lose money
  - Equity holder bears even more risk
- Liquidity (not shown)
  - What if project last not 1 year but 10?

Next time

- **<u>Financial axioms vs</u>**
- **<u>Behavioral finance and imperfect markets</u>**
- **Axioms**
  - **Financial markets are competitive:**
  - **Value additivity;**
  - **No Free Lunch;**
  - **Markets are efficient**
- **Some problems with these axioms**
  - **Market impact**
  - **Behavioral finance**